Name: _____

TOPIC TEST 2

Chapter 9: Directed graphs and networks

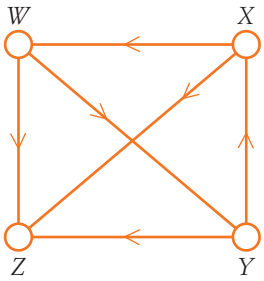
- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 6 short-answer questions (26 marks)
- Part C: 2 analysis questions (20 marks)

Part A

Multiple-choice questions

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 In the digraph shown on the right, a two-stage pathway exists between X and:



A W only

B Z and Y

C W and Z

D Y only

E Z only

[1 mark]

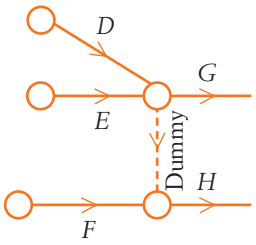
2 The diagram below right shows a section of a network that contains activities D, E, F, G and H. The activity table for these four activities is:

A

Activity	Immediate predecessor
G	D, E
H	F

B

Activity	Immediate predecessor
G	F
H	D, E



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C

Activity	Immediate predecessor
G	*D*, *E*
H	*D*, *E*, *F*

D

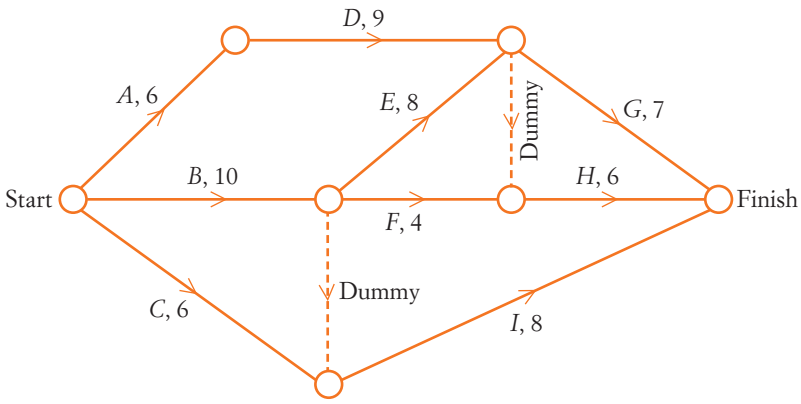
Activity	Immediate predecessor
G	*D*, *E*, *F*
H	*D*, *E*

E

Activity	Immediate predecessor
G	*D*, *E*, *F*
H	*D*, *E*, *F*

[1 mark]

Questions 3 and 4 refer to the information below.
A delivery van is to be repaired. The repair requires every activity in the network below to be completed. Activity times shown are in minutes.



3 The latest starting time for activity *H* is:

A 20 minutes

B 25 minutes

C 14 minutes

D 19 minutes

E 18 minutes

[1 mark]

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4 If activity *H* is delayed by 5 minutes, repair of the van would be delayed by:

A 6 minutes

B 3 minutes

C 5 minutes

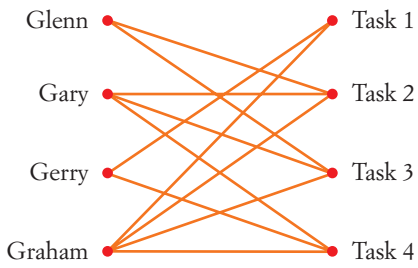
D 0 minutes

E 4 minutes

[1 mark]

5 The bipartite graph shows the tasks that Glenn, Gary, Gerry and Graham are trained to complete.

If each person must be allocated one task, a valid allocation is:



A

Glenn	Task 3
Gary	Task 2
Gerry	Task 4
Graham	Task 1

B

Glenn	Task 1
Gary	Task 2
Gerry	Task 3
Graham	Task 4

C

Glenn	Task 4
Gary	Task 3
Gerry	Task 2
Graham	Task 1

D

Glenn	Task 3
Gary	Task 1
Gerry	Task 4
Graham	Task 2

E

Glenn	Task 3
Gary	Task 4
Gerry	Task 2
Graham	Task 1

[1 mark]

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Part B

Short-answer questions

6 short-answer questions
26 marks
Show your working where appropriate.

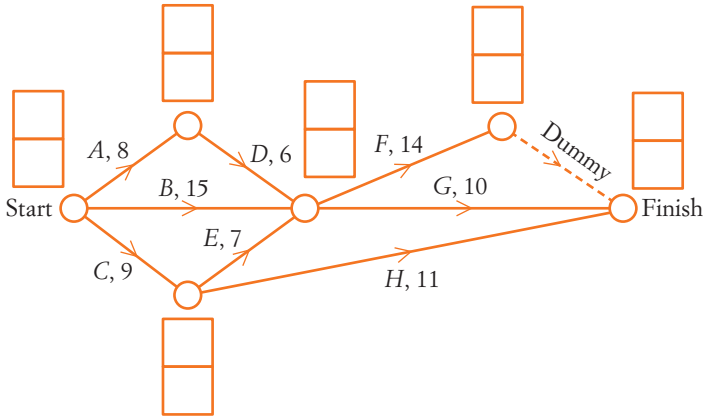
1 Draw the network for the activity table below including dummy activities where required.

Activity	Time (hours)	Immediate predecessor/s
A	5	None
B	5	None
C	6	*A*
D	8	*A*, *B*
E	8	*A*, *B*
F	7	*C*
G	7	*C*
H	3	*D*, *G*
I	6	*E*

[5 marks]

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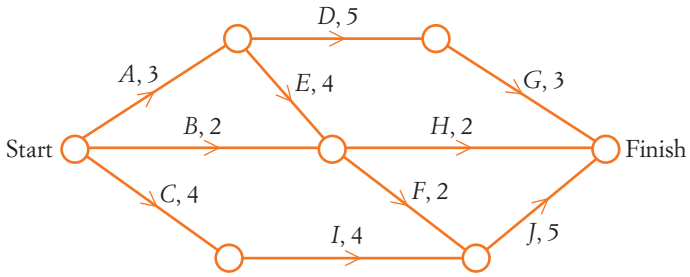
2 Calculate the earliest starting times and the latest starting times for each activity in the network below. Activity times shown are in hours.



[6 marks]

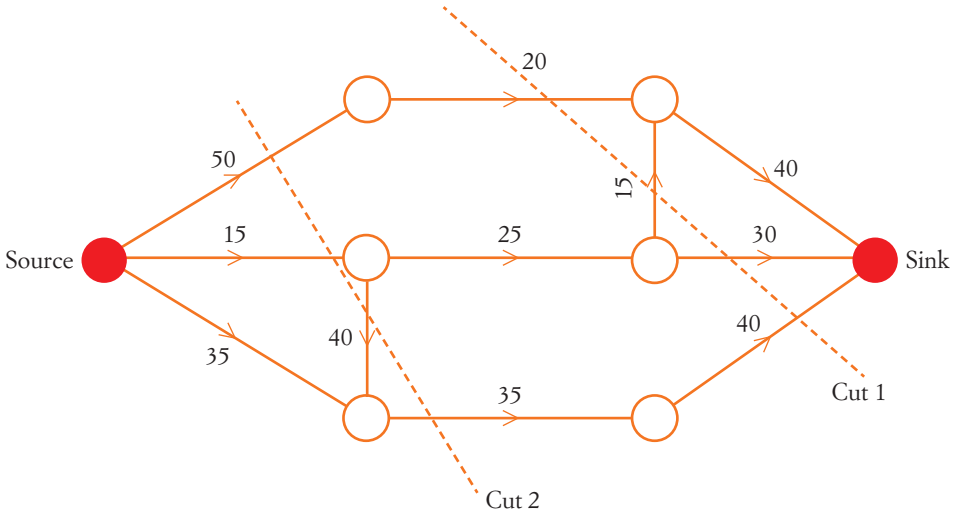
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3 Determine the critical path and the project completion time for the following network. Activity times shown are in days.



[3 marks]

4 The flow capacity, in cars per minute, for a series of one-way roads is shown in the directed graph below.



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Find:

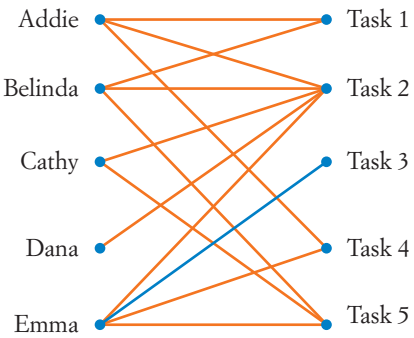
a the flow capacity of cut 1

b the flow capacity of cut 2

c the maximum flow in cars per minute from source to sink.

[4 marks]

5 Determine the allocation of the five tasks for Addie, Belinda, Cathy, Dana and Emma if each person must complete a different task.



[3 marks]

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6 Perform row and column reductions on the matrix that represents the time in minutes for Workers *A*, *B*, *C* and *D* to complete Tasks 1, 2, 3 and 4. Hence, determine the optimum allocation and the minimum time.

Worker	Task 1	Task 2	Task 3	Task 4
A	11	12	13	14
B	10	14	9	11
C	10	9	13	9
D	15	10	12	11

[5 marks]

Part C

Analysis questions

Two analysis questions
20 marks
Show your working where appropriate.

1 A furniture store must arrange deliveries to four customers. The store contracts its deliveries to four different drivers. The distance in kilometres that each driver will need to travel to complete the deliveries is shown in the table below.

	Driver 1	Driver 2	Driver 3	Driver 4
Customer *A*	17	22	17	19
Customer *B*	19	20	20	17
Customer *C*	21	24	23	18
Customer *D*	21	18	20	19

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The deliveries will be allocated in order to minimise the total distance that must be travelled. The Hungarian algorithm is to be used to find this minimum value.

a Step 1 of the Hungarian algorithm is to subtract the minimum entry in each row from each element in the row.

Complete step 1 for Customer *C* by writing the missing values in the table below.

	Driver 1	Driver 2	Driver 3	Driver 4
Customer *A*	0	5	0	2
Customer *B*	2	3	3	0
Customer *C*				
Customer *D*	3	0	2	1

b Complete the Hungarian algorithm so that an allocation can be made.

c Draw the bipartite graph for the final matrix.

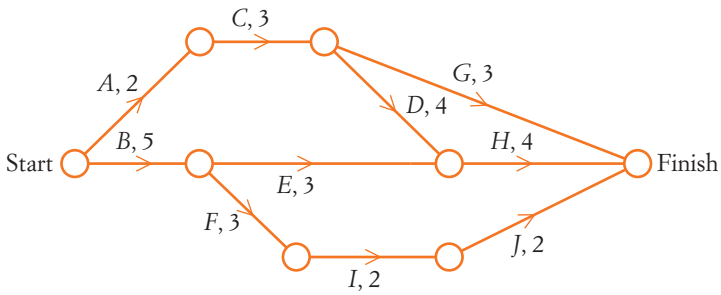
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d Determine the allocation of drivers to customers.

e Find the total distance travelled by the drivers.

[10 marks]

2 A couple decides to remove an old shed from their block and build a new garage. The 10 activities that need to be completed for this project are shown in the network below where activity times are in weeks.



a Find the minimum time required to complete the project.

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b Find the float time for activity *E*.

c Determine the impact on the project completion time if:

i activity *C* is delayed by 1 week

ii activity *G* is delayed by 2 weeks

iii activity *F* is delayed by 2 weeks

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iv activity *B* is delayed by 3 weeks.

The owners are prepared to pay an additional cost to reduce the completion time. The cost of reducing time in each activity is $1500 per week; however, each activity may only be reduced by 1 week.

d Determine the cost of reducing the project completion time by the maximum number of weeks.

[10 marks]

[Total marks: 51]

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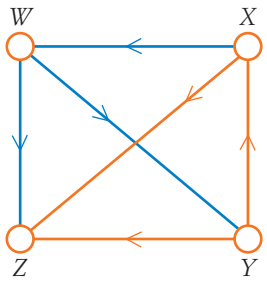
Answers

Part A – Multiple-choice questions [1 mark each]

1 A two-stage pathway is a path between two vertices along two edges.

- Two-stage pathways are XWY and XWZ.
- Two-stage pathway is from X to Y and Z.

$\therefore$ **B**

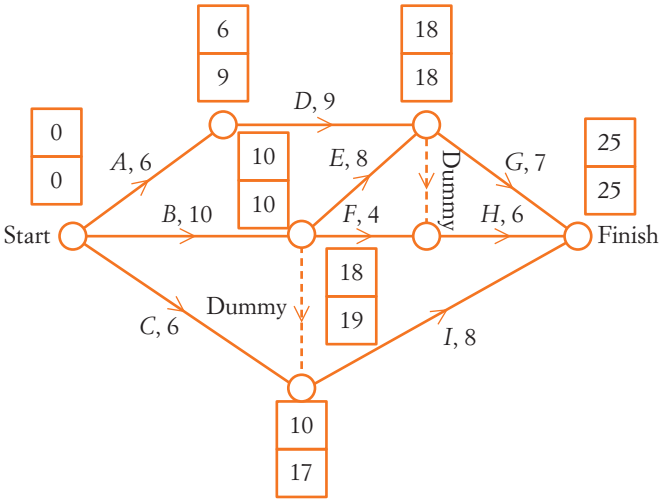


2

Activity	Immediate predecessor
G	D, E
H	D, E, F

$\therefore$ **C**

3



The latest starting time for activity H is 19 minutes.

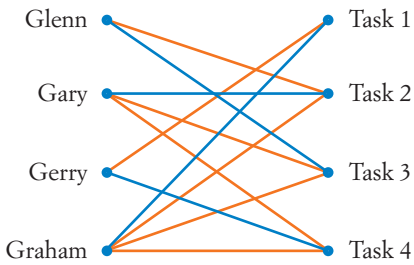
$\therefore$ **D**

4 Activity H has a float of $25 - 18 - 6 = 1$ minute. A delay of 5 minutes will add 4 minutes to the project.

$\therefore$ **E**

5

Glenn	Task 3
Gary	Task 2
Gerry	Task 4
Graham	Task 1

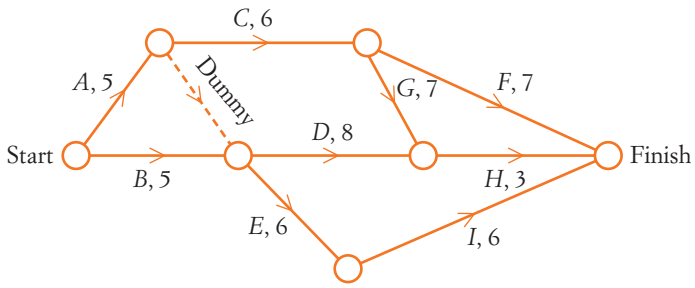


$\therefore$ **A**

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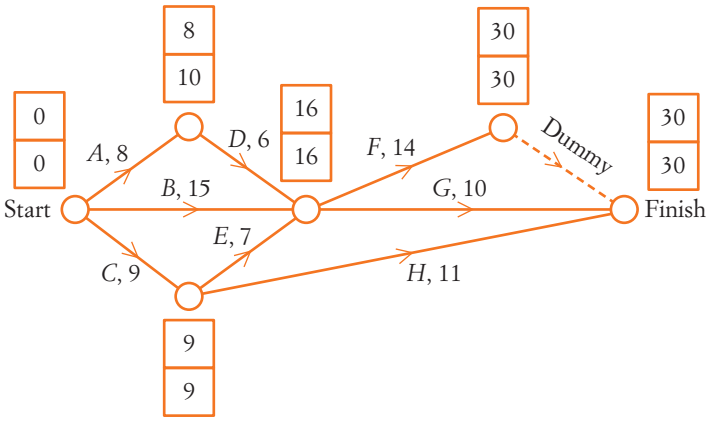
Part B – Short-answer questions

1



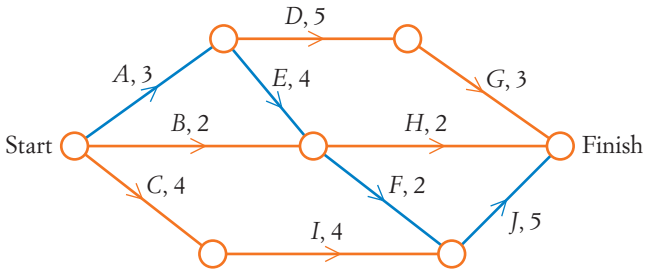
[5 marks]

2



[6 marks]

3 Critical path: *AEFJ*



Project completion time = 3 + 4 + 2 + 5 = 14 days

[3 marks]

4 a Flow capacity of cut 1 = 20 + 15 + 30 + 40

= 105 cars per minute

b Flow capacity of cut 2 = 50 + 15 + 35

= 100 cars per minute

c Maximum flow capacity = capacity of the minimum cut

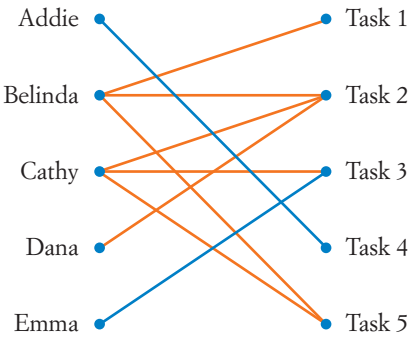
= 20 + 15 + 35

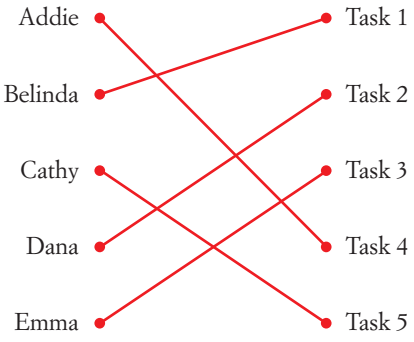
= 70 cars per minute

[4 marks]

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5





- Emma must do Task 3 as she is the only person qualified.
- The other lines are now removed from Emma.
- This leaves Addie as the only person qualified for Task 4.
- Remove the other lines from Addie.
- This means Belinda is the only person qualified for Task 1.
- The process is continued.

Addie	Task 4
Belinda	Task 1
Cathy	Task 5
Dana	Task 2
Emma	Task 3

[3 marks]

6 Original matrix

$$\begin{bmatrix} 11 & 12 & 13 & 14 \\ 10 & 14 & 9 & 11 \\ 10 & 9 & 13 & 9 \\ 15 & 10 & 12 & 11 \end{bmatrix}$$

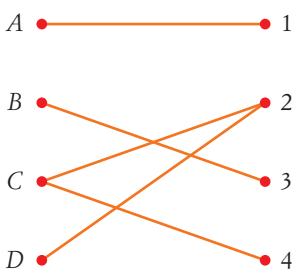
Row reduction

$$\begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 5 & 0 & 2 \\ 1 & 0 & 4 & 0 \\ 5 & 0 & 2 & 1 \end{bmatrix}$$

Column reduction

$$\begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 5 & 0 & 2 \\ 1 & 0 & 4 & 0 \\ 5 & 0 & 2 & 1 \end{bmatrix}$$

Bipartite graph:



Optimum allocation: $A1$, $B3$, $C4$, $D2$

Minimum time: $= 11 + 9 + 9 + 10 = 39$ minutes

[5 marks]

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Part C – Analysis question

1 a

	Driver 1	Driver 2	Driver 3	Driver 4
Customer *A*	0	5	0	2
Customer *B*	2	3	3	0
Customer *C*	3	6	5	0
Customer *D*	3	0	2	1

b Original matrix

$$\begin{bmatrix} 17 & 22 & 17 & 19 \\ 19 & 20 & 20 & 17 \\ 21 & 24 & 23 & 18 \\ 21 & 18 & 20 & 19 \end{bmatrix}$$

Row reduction

$$\begin{bmatrix} 0 & 5 & 0 & 2 \\ 2 & 3 & 3 & 0 \\ 3 & 6 & 5 & 0 \\ 3 & 0 & 2 & 1 \end{bmatrix}$$

Column reduction

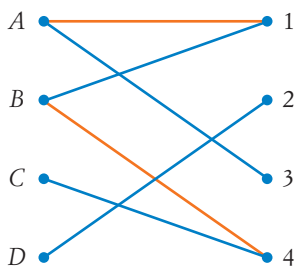
$$\begin{bmatrix} \cancel{0} & 5 & \cancel{0} & \cancel{2} \\ 2 & 3 & 3 & \cancel{0} \\ 3 & 6 & 5 & \cancel{0} \\ \cancel{3} & \cancel{0} & \cancel{2} & \cancel{1} \end{bmatrix}$$

Hungarian algorithm

$$\begin{bmatrix} 2 & 7 & 2 & 6 \\ 2 & 3 & 3 & 2 \\ 3 & 6 & 5 & 2 \\ 5 & 2 & 4 & 5 \end{bmatrix}$$

$$\begin{bmatrix} \cancel{0} & 5 & \cancel{0} & 4 \\ 0 & 1 & 1 & 0 \\ \cancel{1} & 4 & 3 & \cancel{0} \\ \cancel{3} & 0 & 2 & 3 \end{bmatrix}$$

c Bipartite graph



d Allocation of drivers to customers: *A*3, *B*1, *C*4, *D*2

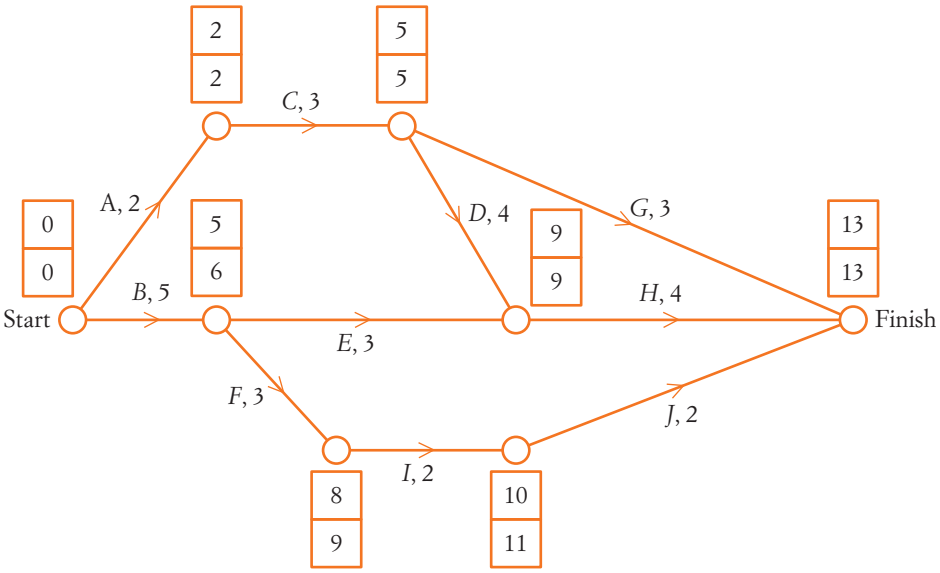
e Distance travelled by the drivers:

$= 17 + 19 + 18 + 18 = 72$ km

[10 marks]

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2



a Critical path: *ACDH*

Project completion time is 13 weeks.

b Float for activity *E*: $9 - 5 - 3 = 1$ week.

c **i** If activity *C* is delayed by 1 week:

activity *C* is a critical activity, so the project is delayed by 1 week.

ii If activity *G* is delayed by 2 weeks:

float for activity *G*: $13 - 5 - 3 = 5$ weeks. Therefore, there is no impact on the project.

iii If activity *F* is delayed by 2 weeks:

float for activity *F*: $9 - 5 - 3 = 1$ week.

Therefore, the project will be completed in $13 + 1 = 14$ weeks.

iv If activity *B* is delayed by 3 weeks:

float for activity *B*: $6 - 0 - 5 = 1$ week.

This delay will mean *BFIJ* and *BEH* are now the critical paths.

Project completion time: $8 + 3 + 4 = 15$ weeks.

d Reduce all activities on the critical path, *A*, *C*, *D* and *H*, by 1 week and also reduce activities *B*, *E*, *F* and *I* by 1 week. This will reduce the project time to 9 weeks.

Cost $= 8 \times \$1500 = \$12\,000$.

[10 marks]

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